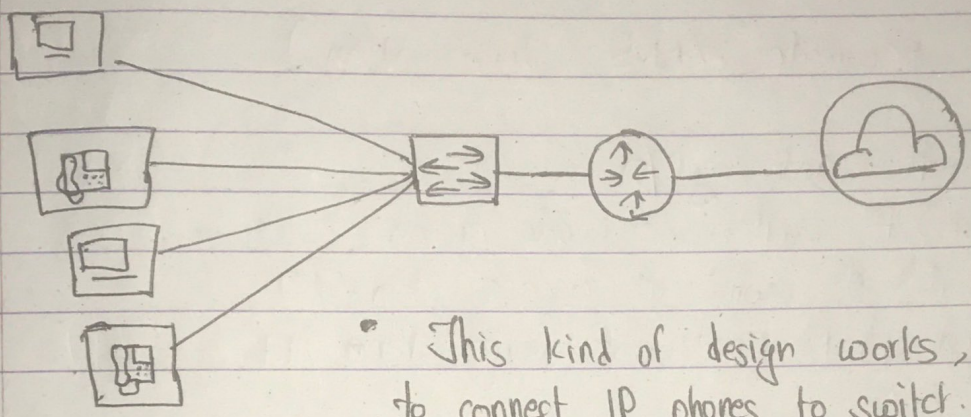
Day-46 August-8

Quality of Service (QOS) PART-I

IP Phones

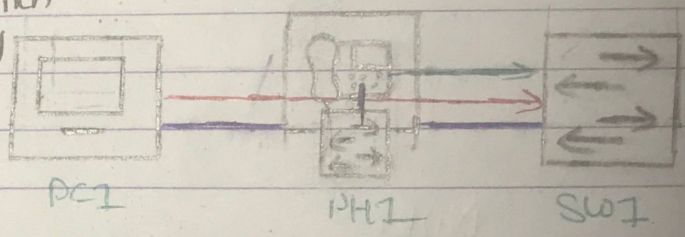
- Traditional phones operate over the public switched telephone networks (PSTN), sometimes also called POTS (Plain old Telephone Service)
- IP phones uses ~~VO~~ VoIP (Voice Over IP) technologies to enable phone calls over an IP network, such as Internet
- IP phones are connected to switch (like other end hosts)



- This kind of design works, but we've better approach to connect IP phones to switch.

[Blue line = 3 port switch]

- IP phones have an internal 3-port switch
  - 1 port connects to external switch (uplink)
  - 1 port connects to PC (downlink)
  - 1 port connects to phone itself.



This setup allows the PC and the IP phone to share a single switch port. ~~Traphic~~ Traffic from PC passes through IP phone to switch (as indicated by red line) Traffic from IP phone can also go to switch (green line).

So with same amount of devices (PC & IP phones) we can use half the no. of switchports (of switch). So we need less switches & we ~~do~~ its ~~ee~~ budget-friendly.

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It is recommended to separate 'voice traffic (from IP phone) and 'data traffic' (from PC) by placing them in separate VLANs.

→ This can be done by using a voice VLAN.

→ Traffic from PC will be untagged, but traffic from phone will be tagged with VLAN id.

Configuring separate VLANS (Voice Vlans)

```
SW1(config)# int g0/0
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 10
SW1(config-if)# switchport voice vlan 11
```

```
SW1(config)# do show int g0/0 switchport
- - - - Name: Gi0/0
- - - - Switchport: Enabled
- - - - Voice VLAN: 11
```

NOB:

i) Another name for access port is untagged port. PC1 will send traffic untagged, as a normal. SW1 will use CDP (Cisco Discovery Protocol) to tell PH1 to tag PH1's traffic in VLAN 11 (send/receive)

ii) Although the interface sends & or receives traffic from two VLANs, its not considered trunk port. It is considered access port

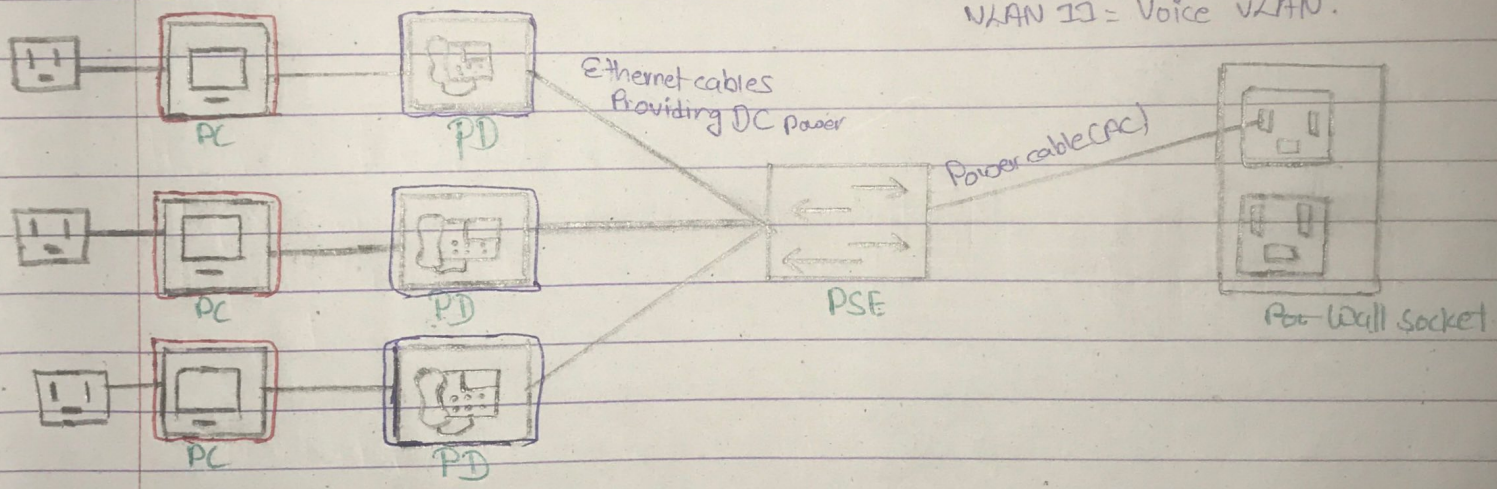
```
SW1(config)# do show interfaces trunk
- - - - - -
```

IP phone is electronic device, so it we need to be connected to electricity to ~~in order~~ in order to work. So we need to connect it to wall socket or some other power outlet to get electricity.

Power over Ethernet (PoE).

- PoE allows power sharing equipment (PSE) to provide power to Powered Devices (PD) over an Ethernet cable.
- Typically the PSE is a switch and PDs are IP phones, IP cameras, wireless access points etc.
- One cable is ~~to~~ used for ~~provide~~ both power as well as to transmit data purpose to
- The PSE receives AC power from outlet, converts it to DC power, and supplies that DC power to the PDs.

VLAN 10 = Data VLAN
VLAN 11 = Voice VLAN.



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